

Sowmya Racha

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Skills

jdhuwh, wkjeiw, kejrle, kwhriw, whreuw

Work Experience

Rich Products Corporation | Remote, USA

Jan 2025 - Mar 2026

Software Engineer – AI

- Developed scalable RESTful APIs and microservices using Java, Spring Boot, and Kafka in a distributed microservices architecture, supporting high-volume product and order processing workflows. Implemented OAuth2/JWT authentication and role-based access control (RBAC) using Spring Security, securing 20+ APIs across multiple services. Designed event-driven architecture using Apache Kafka, enabling real-time inventory and order processing across distributed systems. Deployed cloud-native applications on AWS (EC2, RDS, S3) using Docker and implemented CI/CD pipelines using Jenkins, reducing deployment time by 40

USAA- American Insurance and Financial Company | Hyderabad, India

Aug 2021 - Feb 2024

Fullstack Software Engineer

- Developed and maintained scalable microservices and REST/GraphQL APIs processing 10M+ daily transactions using Node.js, PostgreSQL, and MongoDB. Improved backend performance by 25

Quesite | Hyderabad, India

Aug 2020 - May 2021

Software Developer

- Developed enterprise web applications using AngularJS, Core Java, and REST APIs for employee management and payroll systems. Implemented client-side validation and exception handling, reducing data entry errors by 20

Project Work

Automatic resume generator

Scalable Machine Learning Solution Development

I independently developed a comprehensive machine learning solution for a high-stakes problem using Python, Scikit-learn, and TensorFlow. I crafted a robust data pipeline to preprocess, engineer, and train models, yielding significant accuracy gains through model tuning, validation, and ensemble methods. Leveraging Docker and FastAPI, I successfully deployed the model in a production-ready AWS environment. By monitoring model drift, latency, and reliability, I ensured seamless integration with cross-functional Agile teams. This project showcases my expertise in designing, deploying, and maintaining scalable machine learning solutions.

AI agent for diet

"Developed and deployed a suite of predictive models, including Word Token Regression, Word Token Logistic Regression, and other custom algorithms, using Python, Scikit-learn, TensorFlow, and PyTorch. Designed and optimized data pipelines for efficient preprocessing, feature engineering, and model training. Successfully deployed models on AWS/GCP environments using Docker and FastAPI, ensuring high reliability and scalability. This personal

project showcases my ability to work independently, design end-to-end machine learning solutions, and integrate with DevOps tools, demonstrating my expertise as a Machine Learning Engineer."

Automatic YouTube Shorts generator

Developed an automatic YouTube short video generator using AI. The content, images, and voice are generated using LLMs, Diffusion models, and text-to-speech models from the Huggingface hub.

Breast Cancer Detection

Used Pytorch to train extended VGG11 vision model on CBIS-DDSM dataset to classify breast cancer into 3 classes. After training on A100 GPU for few hours, the trained model accuracies were Train Accuracy: 71.34% , Validation Accuracy: 74.30%.

Virtual Reality based Earth/Moon explorer

Developed a Virtual Reality (VR) application in Unity 3D using C# to explore Earth/Moon in VR. Used REST API to get geospatial data (height maps) of Earth and the Moon from usgs.gov, and converted the geospatial data into 3D models within the application. People can visit almost any place on Earth and the Moon in VR.

Diffusion model to generate faces

Created a diffusion model using the Huggingface Diffusers library. Trained a UNet2D model from scratch on the MetFaces dataset using Google Colab Pro to generate human faces resembling works of art.

Automatic Job applications

I am using LLMs and webdriver to apply to jobs automatically. The LLM reads the source HTML and screenshot of the webpage to identify what needs to be filled. I am sending random keys into text fields to identify which text field corresponds to which information. (for example, I send xyz to "first name" field, I ask the LLM what is the text within "first name" field, based on that we can map the input field to 'first name' dictionary entry, which contains my first name, I send the correct key now). Predefined questions are given to LLM and I process the responses to fill my details in a job application.

Education

University of Texas at Arlington | Texas, USA

Jan 2024 - Dec 2025

M.S. in Mathematics and Computer Science GPA: 3.9/4.0 Specialty areas: Big Data and Machine Learning

Srinidhi Institute of Science and Technology | Hyderabad, India

Aug 2017 - May 2021

B.Tech in Computer Science